Grade 7
Unit 4
Lesson 2 Practice

Name Answers
Date
Period

1. The formula to find the area of a trapezoid is

- ○ $\frac{1}{2}d_1d_2$
- ○ bh
- ● $\frac{1}{2}h(b_1 + b_2)$

2. The formula to find the area of a rhombus is

- ● $\frac{1}{2}d_1d_2$
- ○ bh
- ○ $\frac{1}{2}h(b_1 + b_2)$

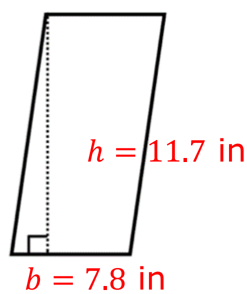
3. The formula to find the area of a parallelogram is

- ○ $\frac{1}{2}d_1d_2$
- ● bh
- ○ $\frac{1}{2}h(b_1 + b_2)$

For questions 4-6, label each figure with the given dimensions, then calculate the area using the appropriate formula.

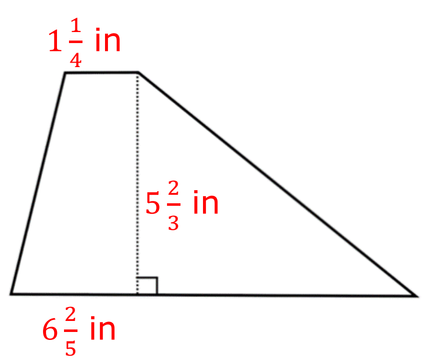
4. b = 7.8 in
h = 11.7 in
$7.8 \cdot 11.7$

Area = 91.26 in²



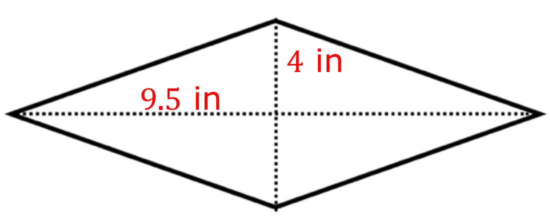
5. $b_1 = 1\frac{1}{4}$ in
$b_2 = 6\frac{2}{5}$ in
$h = 5\frac{2}{3}$ in
$\left(1\frac{1}{4} + 6\frac{2}{5}\right) = 7\frac{13}{20}$ or 7.65
$\frac{1}{2} \cdot 5\frac{2}{3} \cdot 7.65$

Area = $21\frac{37}{40}$ or 21.675 in²



6. $d_1 = 9.5$ in
$d_2 = 4$ in
$\frac{1}{2} \cdot 9.5 \cdot 4$

Area = 19 in²



7. Jocelyn made an error in her work when solving the problem below. Identify the error and then find then find the base of the parallelogram.

The area of a parallelogram is 75 square centimeters and the height is 6.25 centimeters. What is the length of the base of the parallelogram?

Her solution: $(75)(6.25) = 468.75$ → error is she multiplied the area by the height to find the base

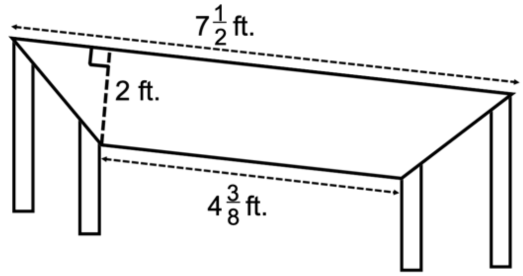
The length of the base is 468.75 centimeters

Area= $b \cdot h$

$\frac{75}{6.25} = \frac{6.25b}{6.25}$

$b = 12$ cm

8. The seat of a bench in the Marlins dugout is in the shape of a trapezoid. Find the area of the bench.



$A = \frac{1}{2} \cdot h \cdot (b_1 + b_2)$

$= \frac{1}{2} \cdot 2 \cdot \left(4\frac{3}{8} + 7\frac{1}{2}\right)$

$= \frac{1}{2} \cdot 2 \cdot 11\frac{7}{8}$

$= 11\frac{7}{8}$ ft²

9. The Adidas Flagship Building located in Herzogenaurach is shaped like a rhombus. One of the diagonals measures 255.4 meters and the other diagnol measures 203.7 meters. Find the area of the top of the building.

$A = \frac{1}{2} \cdot d_1 \cdot d_2$

$= \frac{1}{2} \cdot 255.4 \cdot 203.7$

$= 26012.49$ m²

10. An eraser is in the shape of a parallelogram with a base of $1\frac{7}{12}$ inches and a height of $\frac{6}{14}$ inches. How many square inches is the top of the eraser?

$A = b \cdot h$

$= 1\frac{7}{12} \cdot \frac{6}{14}$

$= \frac{19}{28}$ in²